

KOÇ ÜNİVERSİTESİ

MECH 303 – Project 3

Finite Element Modeling of a Bolted Joint

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Abstract

This project presents a comprehensive computational structural analysis of a bolted lap joint assembly using ANSYS Mechanical. The study aims to investigate the complex interactions between bolt pretension, contact nonlinearity, and external tensile loading to determine the joint's structural integrity and failure limits. The analysis is divided into three distinct phases. In Task 1, the assembly was subjected to a standard external load of 25 kN following a bolt pretensioning step. The Finite Element Analysis (FEA) results revealed significant stress concentrations at the bolt head fillet and confirmed that the joint maintained sufficient contact pressure to prevent leakage or slip under service conditions. Task 2 involved a parametric study to identify the Separation Force, the critical load at which the bracket physically separates from the plate. By incrementally increasing the external load and monitoring the average contact pressure, the separation point was determined to be approximately 45.53 kN. The results showed a linear decay in contact pressure and validated the non-linear stiffness response of the bolted joint after separation. Finally, Task 3 focused on design optimization for a specific service load of 6 kN. A material selection process was conducted, replacing the original high-strength bolt with a cost-effective Metric Class 5.8 bolt. By optimizing the preload to 15,000 N, the design achieved a Safety Factor 1.3 against both yielding and separation, ensuring a safe and efficient joint configuration.

1. Introduction

Bolted joints are widely used in mechanical assemblies to create rigid, detachable connections between structural components. The reliability of such joints depends fundamentally on the relationship between the internal clamping force, generated by bolt pretension, and the external service loads applied to the structure. If the external load exceeds a critical threshold, the joint members may separate, leading to a loss of stiffness, bolt loosening, and eventually fatigue failure. Therefore, understanding the mechanics of pretensioned joints and predicting their behavior under different loads is crucial for ensuring structural safety.

This project focuses on the finite element analysis (FEA) of a single-bolt bracket-plate assembly using ANSYS Mechanical [2]. Unlike simple theoretical calculations, computational simulations allow for a detailed investigation of complex phenomena such as contact pressure distribution, stress concentrations at thread roots, and the non-linear response of the joint during separation. By utilizing Computer-Aided Engineering (CAE) tools, engineers can identify potential failure points and optimize designs before physical prototyping.

The specific objectives of this study are threefold. First, to simulate the pretensioning process and analyze the stress distribution within the bolt and the bracket under a standard operating load. Second, to conduct a parametric study to determine the Separation Force, the precise external load at which the joint loses contact integrity. Finally, to perform a design optimization study to select the most appropriate bolt grade and preload level that satisfies safety factors while minimizing cost and material usage. Through these tasks, the project aims to demonstrate the practical application of FEA in solving real-world structural mechanics problems.

2. Description of the Physical Problem

The physical problem analyzed in this project involves the structural assessment of a bolted lap joint, a fundamental connection type in mechanical engineering. The assembly consists of an L-shaped bracket attached to a rigid base plate using a 4 bolt and nut [1]. The system is subjected to a two-stage loading process: initial tightening of the bolt (pretension) followed by the application of an external tensile load that tends to separate the bracket from the plate. The primary goal is to simulate the non-linear behavior of this joint, including contact status, stress distribution, and the critical separation limit, using the Finite Element Method (FEM).

2.1. Geometry and Model Details

The 3D geometry consists of five components: the bracket, the base plate, the pin, the M12 bolt, and the nut can be seen in Figure 1. To optimize computational efficiency, a Half-Symmetry model was utilized. Since both the geometry and the loading conditions are symmetric with respect to the mid-plane, simulating only half of the model significantly reduces the node count and solution time without compromising accuracy.

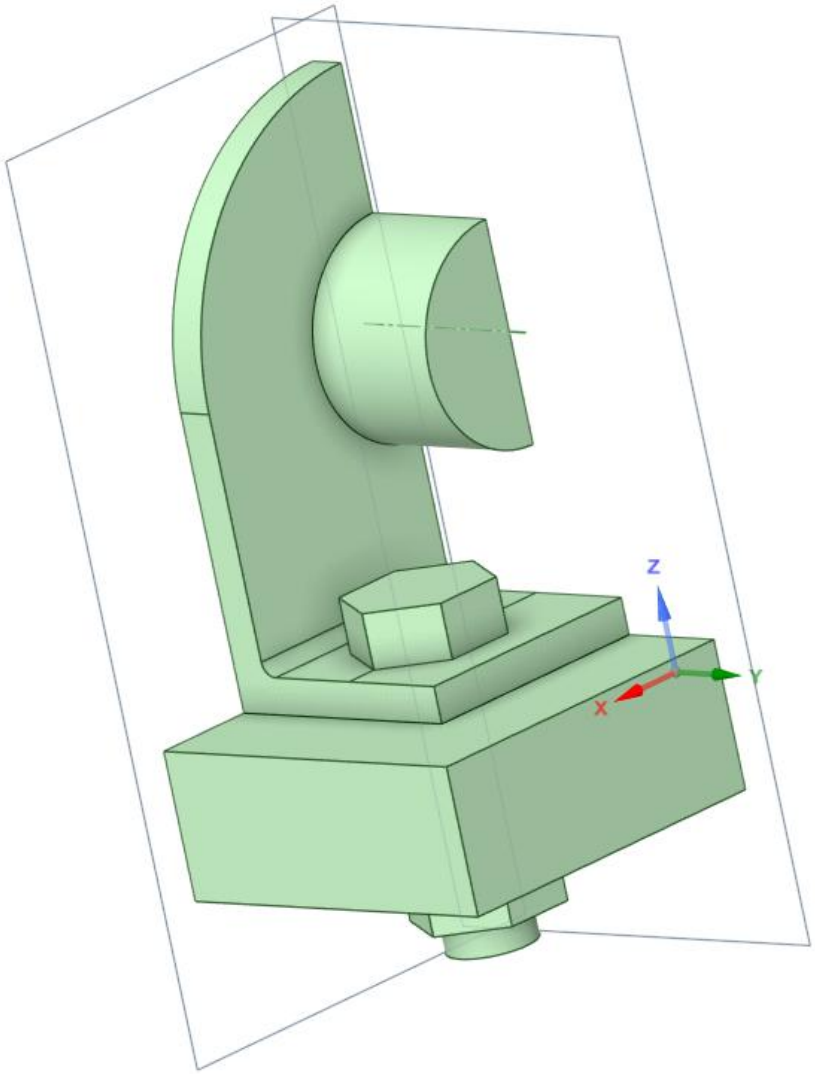


Figure 1: 3D Geometry of the bolted joint assembly modeled with

The interaction between the components is defined using non-linear contact pairs. A Frictional Contact with a friction coefficient of $\mu = 0.2$ is applied at the bracket-to-plate and bolt head-to-bracket interfaces, allowing for realistic sliding and potential separation. A Bonded Contact is used between the nut threads and the bolt shank to simulate a secure threaded connection.

2.2. Material Properties

The material definition is a critical aspect of the simulation, particularly for the optimization phase in Task 3. The structural components (bracket, plate, and pin) are modeled using standard Structural Steel for all analysis tasks. However, the bolt material is varied to meet the specific requirements of each task:

- **Task 1 and Task 2:** The bolt and nut are assigned high-strength steel properties (see Figure 2) to withstand high pretension loads without plastic deformation.

Young's Modulus	200	GPa
Poisson's Ratio	0,3	
Bulk Modulus	1,6667E+11	Pa
Shear Modulus	7,6923E+10	Pa
Tensile Yield Strength	660	MPa

Figure 2: Material properties of SAE 8.8 for task 1 and 2

- **Task 3:** To achieve a cost-effective design for a lower service load (6 kN), the bolt material is changed to Metric Class 5.8 Carbon Steel.

Material Assignment	**Component(s)**	**Density (kg/m3)**	**Young's Modulus (GPa)**	**Poisson's Ratio**	**Yield Strength (MPa)**
Structural Steel	Bracket, Plate, Pin	7850	200	0.3	250
Bolt Class 5.8	Bolt, Nut (Task 3)	7850	200	0.3	400

Table 1: The specific mechanical properties used in the ANSYS Engineering Data are summarized. [3]

2.3. Loading and Boundary Conditions

The simulation is structured into two static time steps to replicate the physical installation and operation sequence. The boundary conditions and loads are applied as follows:

1. **Symmetry:** A frictionless symmetry support is applied to the cut faces of all components Displacement $X = 0$. Figure 3.
2. **Fixed Support:** The side faces of the base plate are fully constrained to simulate a rigid mounting structure.
3. **Bolt Pretension (Step 1):** An internal pretension load is applied to the bolt shank.

- For Task 1 & 2, a high pretension load (approx. 45 kN) is applied to maximize clamping.
- For Task 3, an optimized preload of 15,000 N is applied to prevent yielding of the Class 5.8 bolt.

4. **External Load (Step 2):** The pretension is "Locked," and an external tensile force is applied to the pin face (Figure 4).
    - **Task 1:** A standard load of 25 kN.
    - **Task 2:** A parametric ramp load increasing from 25 kN to 45.53 kN to detect separation.
    - **Task 3:** A service load of 6 kN.

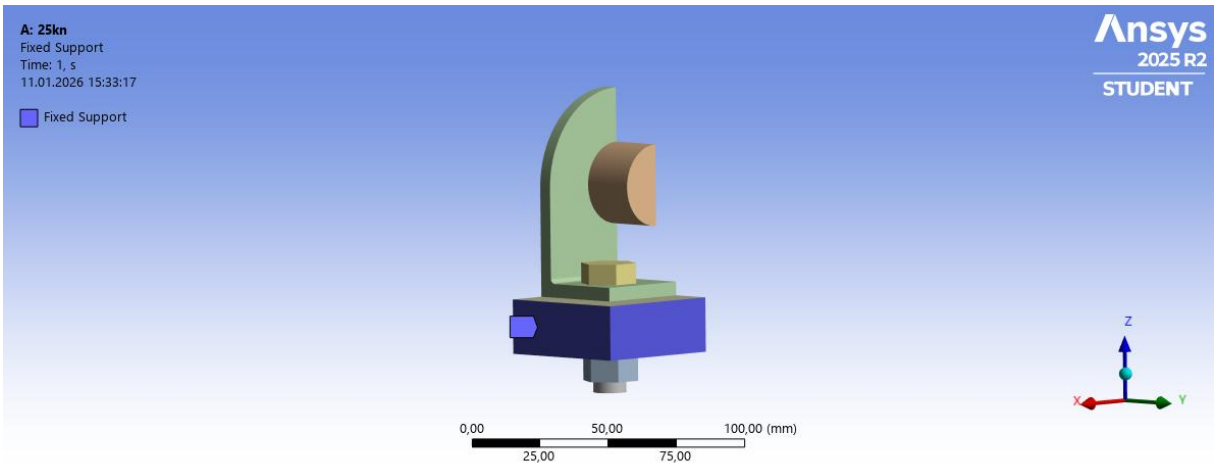


Figure 3: Fixed bodies of the model to prevent it from moving

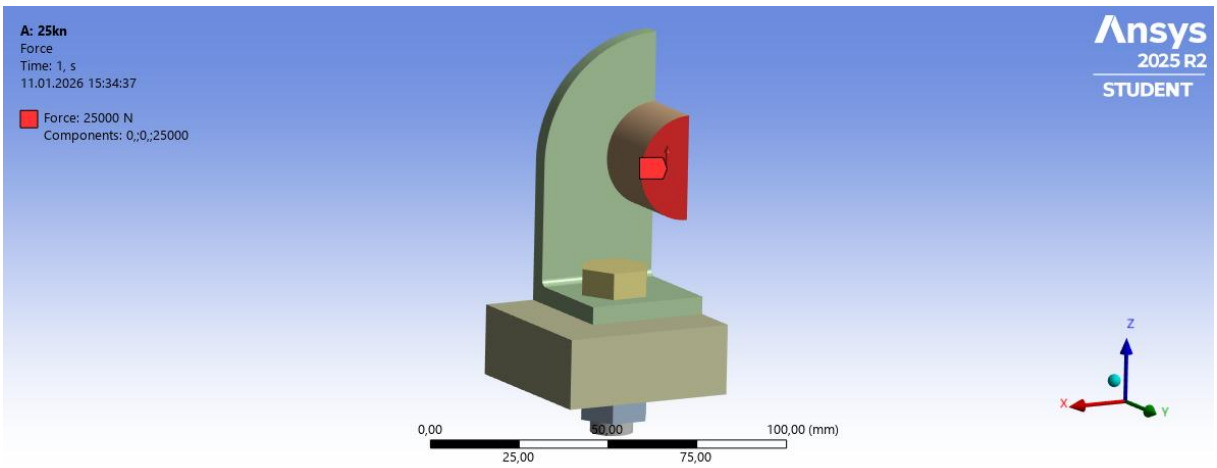


Figure 4: Force applied for task 1

3. Results and Discussions

3.1. Task 1: Analysis under Standard Service Load

In the first step of the analysis, a bolt pretension load was applied to the bolt shank to simulate the clamping force before any external service loads were introduced. The pretension force was calculated based on the formula $F_i = 0.9A_tS_p$ and applied to the model. The simulation with 3mm mesh (Figure 5) results for Step 1 are presented below, detailing the stress distribution, contact behavior, and bolt loads.

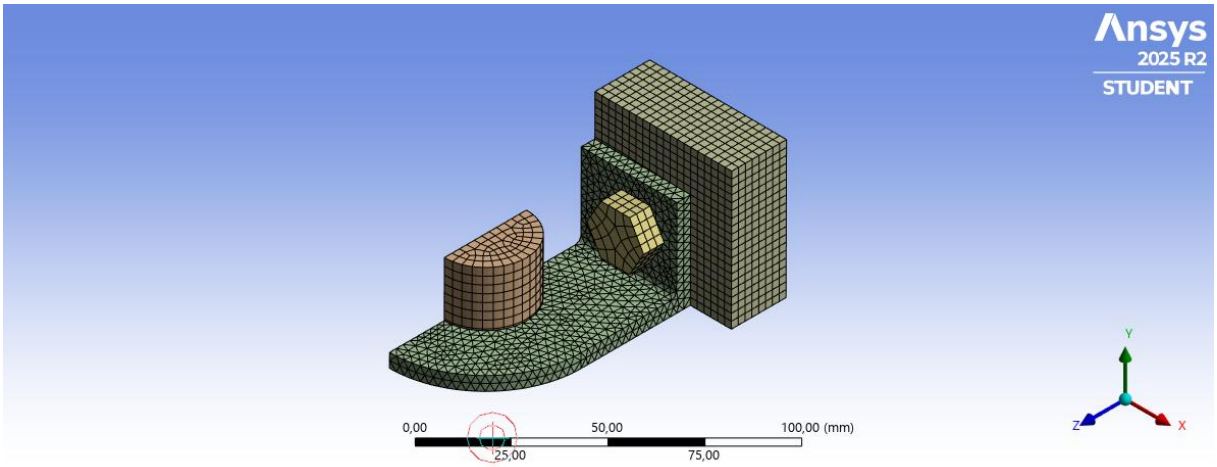


Figure 5: Mesh of the part 3 mm

The equivalent stress distribution on the assembly (bracket, plate, and pin) is shown in Figure 6. The maximum stress concentration is observed at the contact regions between the bolt head/nut and the structural components, reaching a maximum of approximately 626.13 MPa. This local high stress is expected due to the compressive clamping forces exerted by the fastener.

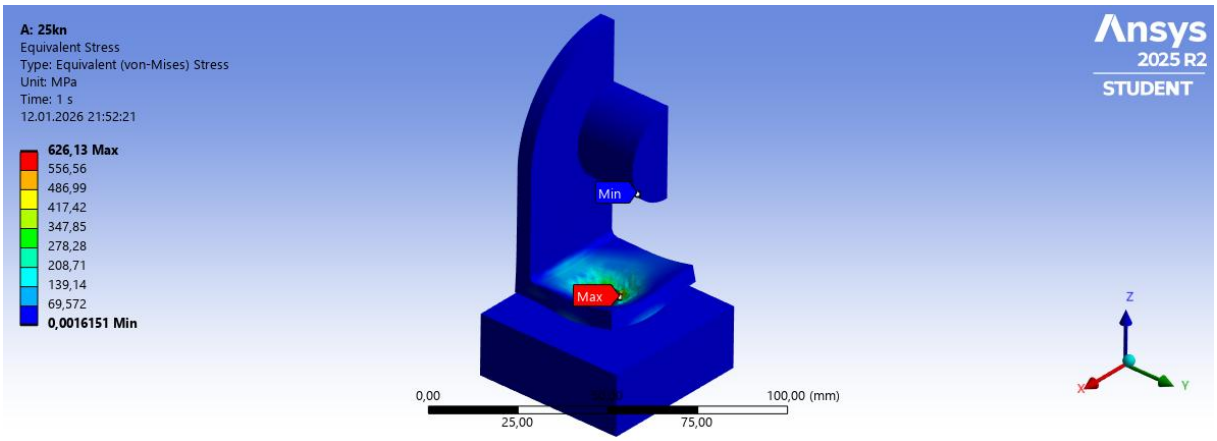


Figure 6: Equivalent (Von-Mises) Stress distribution on the assembly under pretension loading.

The normal stress along the Z-axis (axial direction) of the bolt is illustrated in Figure 7. The bolt shank exhibits a uniform tensile stress distribution, which confirms that the pretension has been applied correctly. The maximum tensile normal stress recorded is 497.28 MPa, which is within the elastic range for 8.8 SAE Class Steel.

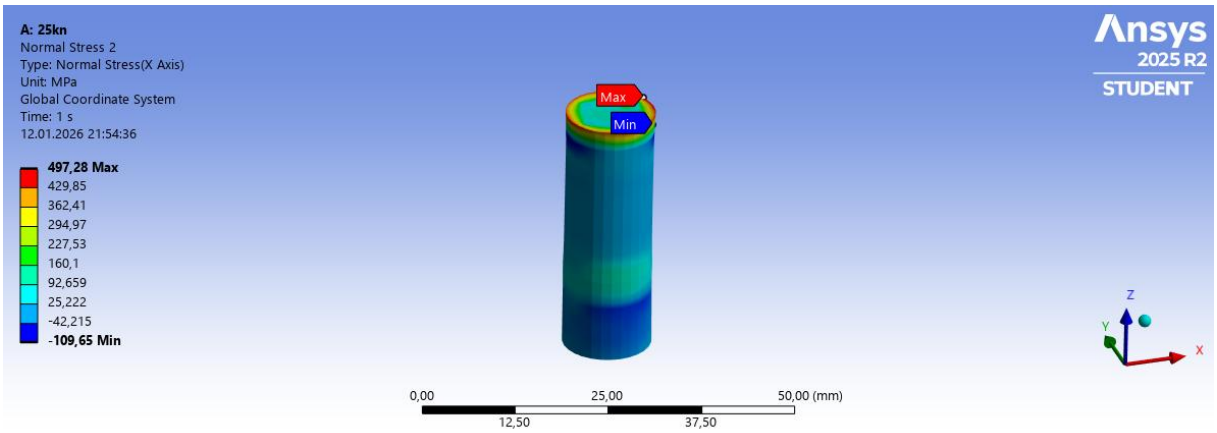


Figure 7: Normal Stress distribution along the bolt shank.

The contact pressure distribution on the bracket's lower face is shown in Figure 8. The pressure is distributed in an annular pattern around the bolt hole, which is characteristic of a bolted joint.

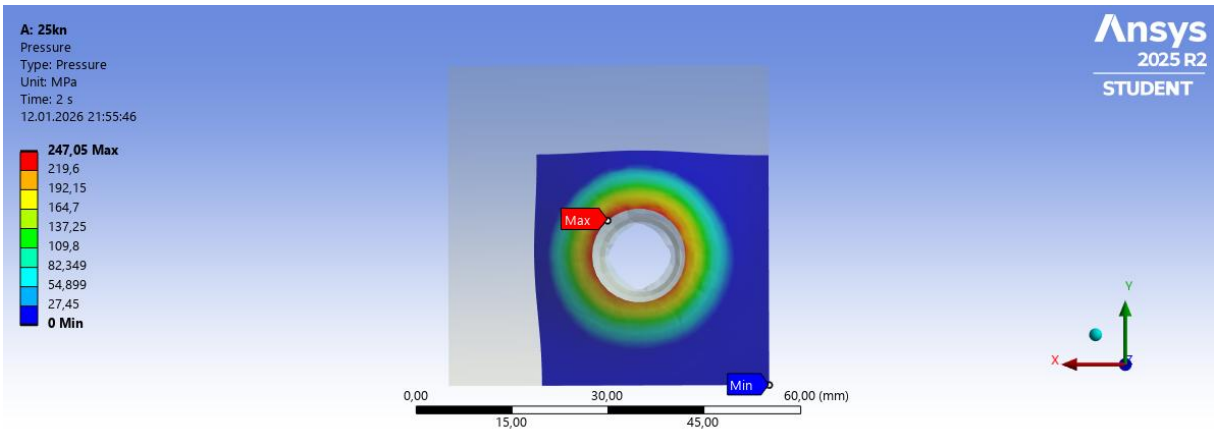


Figure 8: Contact Pressure distribution on the bracket's lower face.

Furthermore, the contact status (Figure 9) indicates that the region directly under the clamping load is in a "Sticking" state (orange zone).

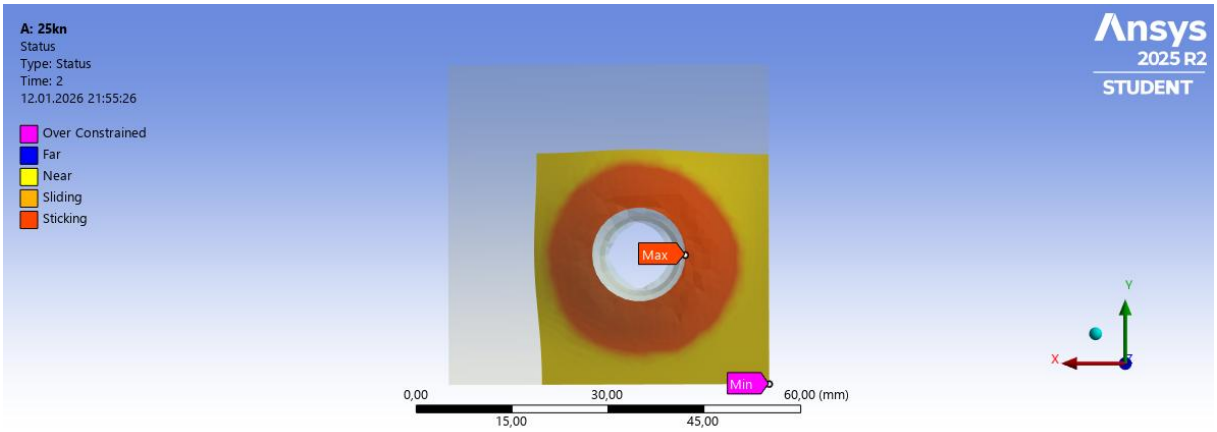


Figure 9: Contact Status on the bracket's lower face.

The resulting adjustment force (Working Load) in the bolt is presented in Figure 10. Picture indicates a locked pretension value of approximately 45.5 kN, which aligns with the calculated preload requirements.

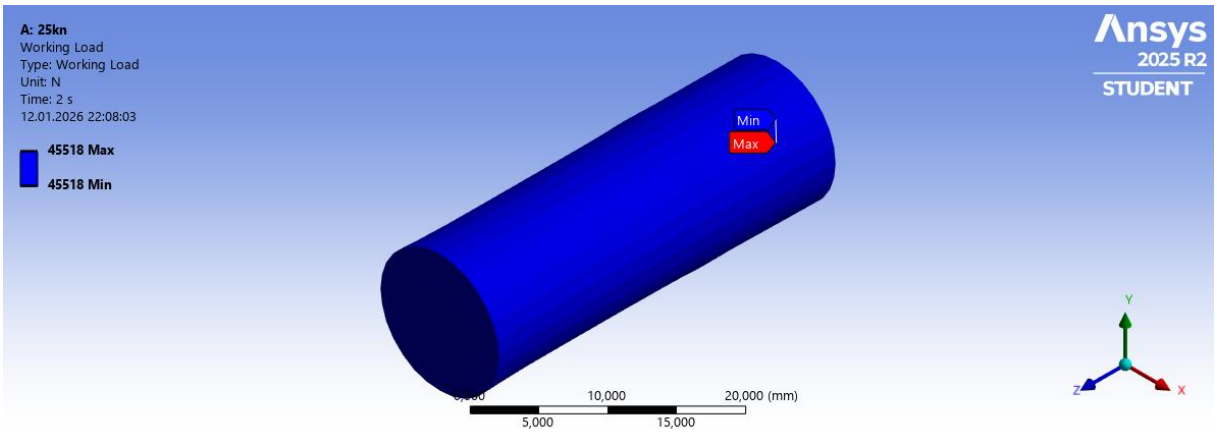


Figure 10: Bolt Working Load after locking pretension.

The bolted assembly was analyzed under a standard external operating load. In the second step, an external tensile load of 25 kN was applied to the pin, creating a separation force on the bracket. The primary objective was to evaluate the stress distribution and verify the contact integrity of the joint under this service load.

The Equivalent (Von-Mises) stress distribution on the bracket and the bolt is presented in Figure 11.

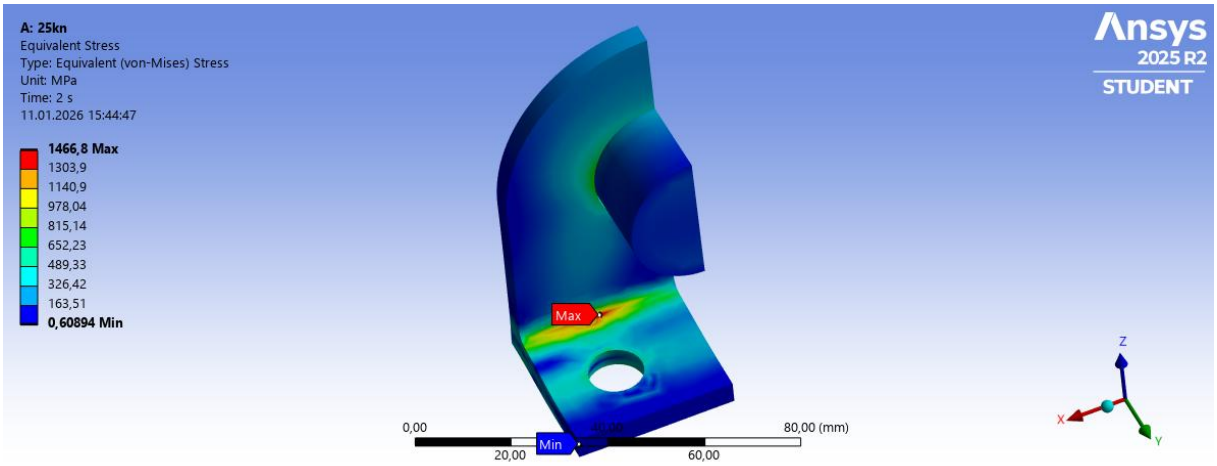


Figure 11: Equivalent stress distribution showing stress concentrations at the bolt head fillet and bracket hole.

As observed in the contour plot, the maximum stress is not located on the bulk material of the bracket but is concentrated at the fillet region between the bolt head and the shank. This local stress concentration is a geometric effect caused by the sharp transition and the high axial tension in the bolt. Furthermore, the bracket exhibits a bending behavior due to the

nature of the loading, often referred to as prying action. This deformation leads to increased tensile stresses on the inner surface of the bracket near the bend radius.

The structural behavior of the bolt itself is quantified by the Working Load, which represents the total axial force carried by the bolt after the external load is applied at Figure 12.

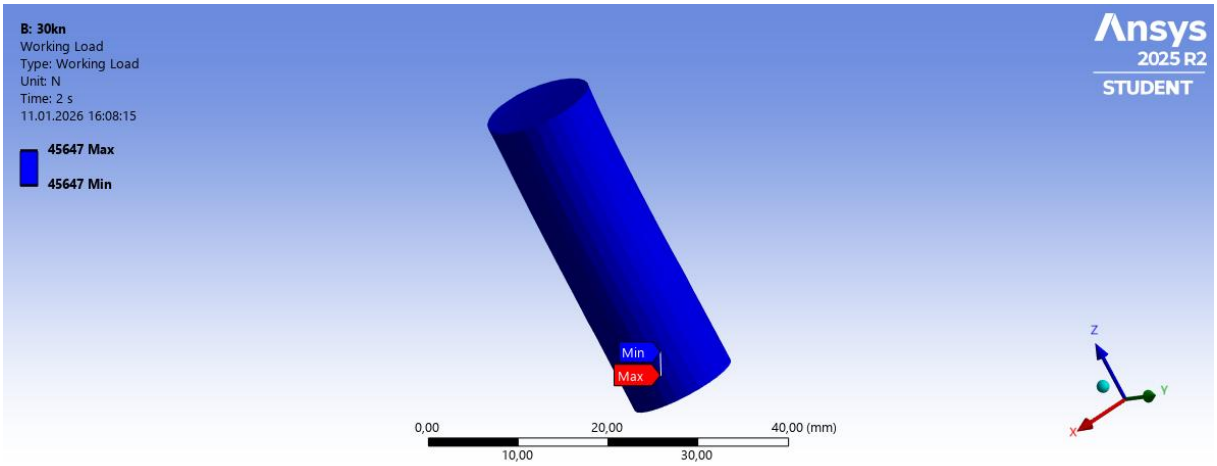
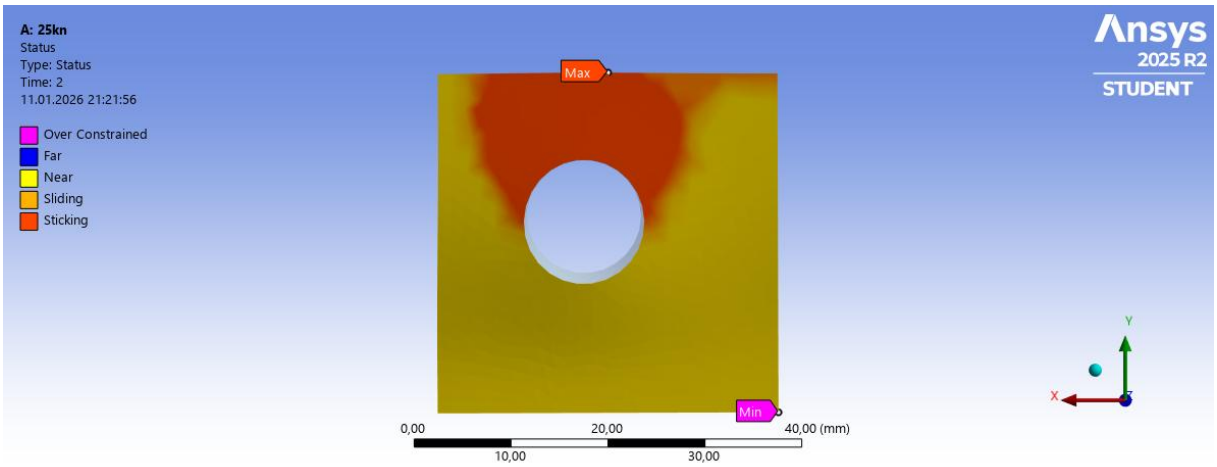


Figure 12: Total axial load on the bolt under 25 kN external force.

The analysis shows that the bolt working load increased slightly from the initial pretension value. However, this increase is minimal because the joint stiffness is relatively high compared to the bolt stiffness. The bolt material remains within the elastic region, and the safety factor against yielding is sufficient for this load case.

The most critical parameter for the joint's functionality is the contact status between the bracket and the base plate. The contact pressure distribution is illustrated in Figure 13.



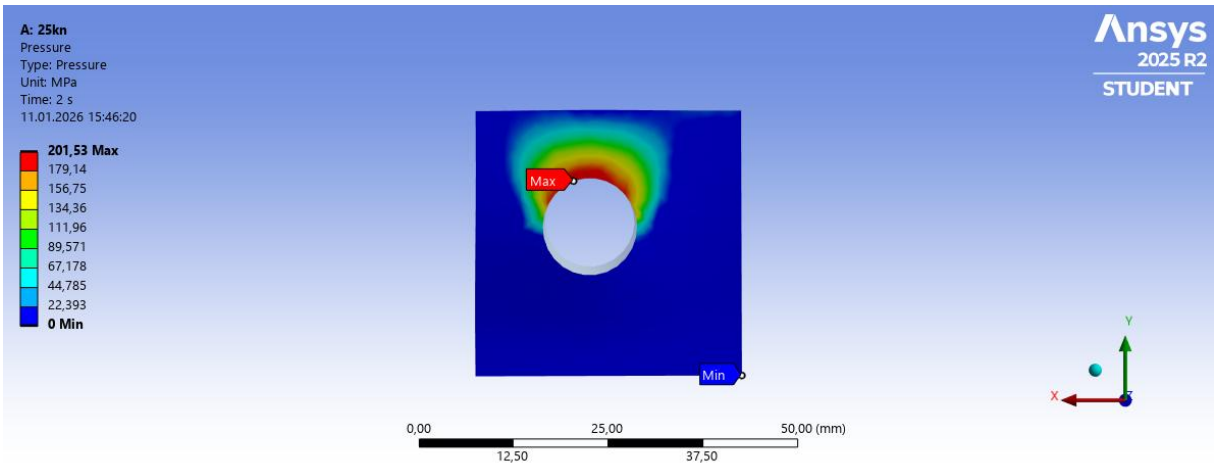


Figure 13: Top is state of the contact plane and at the bottom contact pressure map at the bracket-plate interface under 25 kN load.

The contour plot indicates that the contact pressure is non-zero around the bolt hole, confirming that the clamping force is sufficient to keep the joint closed. The pressure is highest in the immediate vicinity of the bolt hole and decreases radially. Although the edges of the bracket show a tendency to lift off (zero pressure), the effective contact area remains large enough to transmit the shear forces and maintain the structural integrity of the assembly. Therefore, under the 25 kN service load, the joint is considered safe against separation.

In addition to the equivalent stress, the Normal Stress distribution along the bolt was evaluated to understand the specific tensile and bending behaviors.

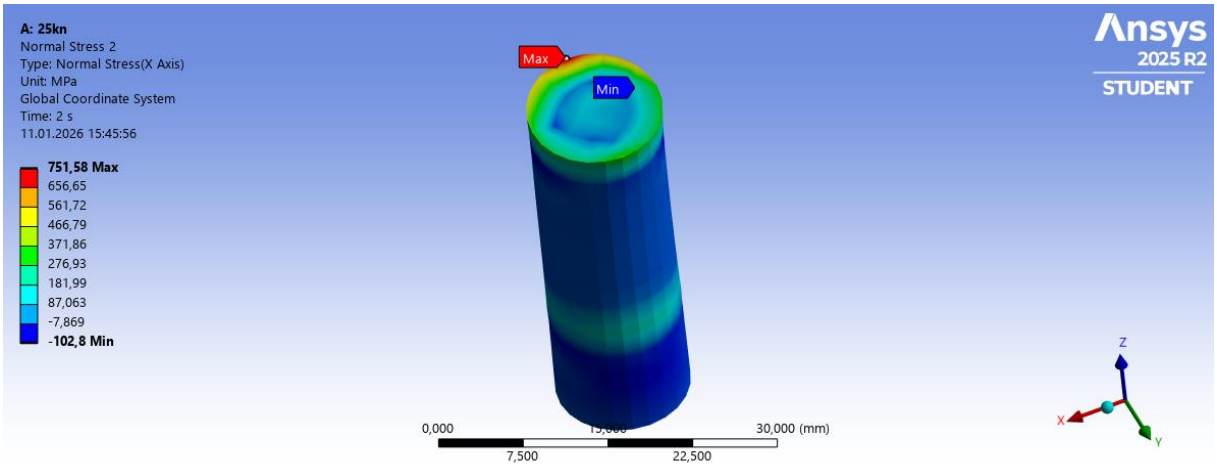


Figure 14: Normal stress distribution on the bolt showing the combination of axial tension and bending.

The presence of a compressive stress region (indicated by negative values, approx. -102 MPa) on one side of the bolt confirms that the bolt is subjected to a significant bending moment. This phenomenon, known as prying action, occurs because the bracket deforms and levers against the plate, imposing a non-axial load on the bolt.

3.2. Task 2: Parametric Study and Separation Analysis

Following the static analysis in Task 1, a parametric study was conducted to investigate the non-linear response of the joint under incrementally increasing external loads. The external tensile load was raised from 25 kN up to the critical separation point. The primary objective was to visualize the gradual loss of contact area and to quantify the relationship between the external force, the bolt working load, and the contact pressure.

3.2.1. Evolution of Contact Pressure

The contact status between the bracket and the base plate is the most critical indicator of joint integrity. To visualize the separation process, contact pressure contours were extracted at intermediate load steps: 30 kN, 35 kN, and 40 kN.

At 30 kN, the contact pressure distribution remains similar to the standard service load case. The high-pressure zone around the bolt hole effectively maintains the seal, although a slight reduction in the contact area at the bracket edges is observed.

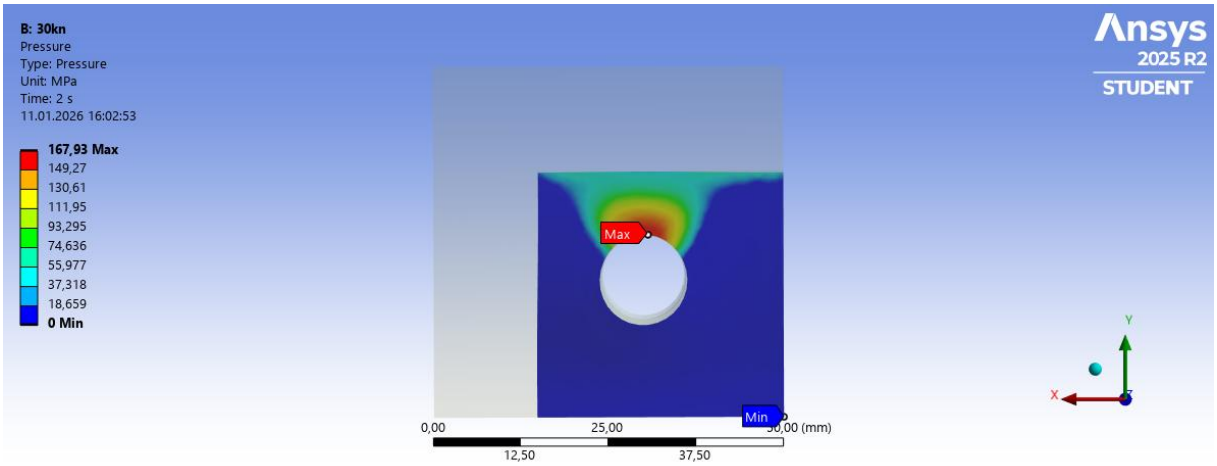


Figure 15: Contact pressure distribution at 30 kN external load.

As the load increases to 35 kN at Figure 16, the "lifting" effect becomes more pronounced. The contact region begins to shrink towards the center of the bolt hole, indicating that the bracket is flexing and pulling away from the plate at the periphery.

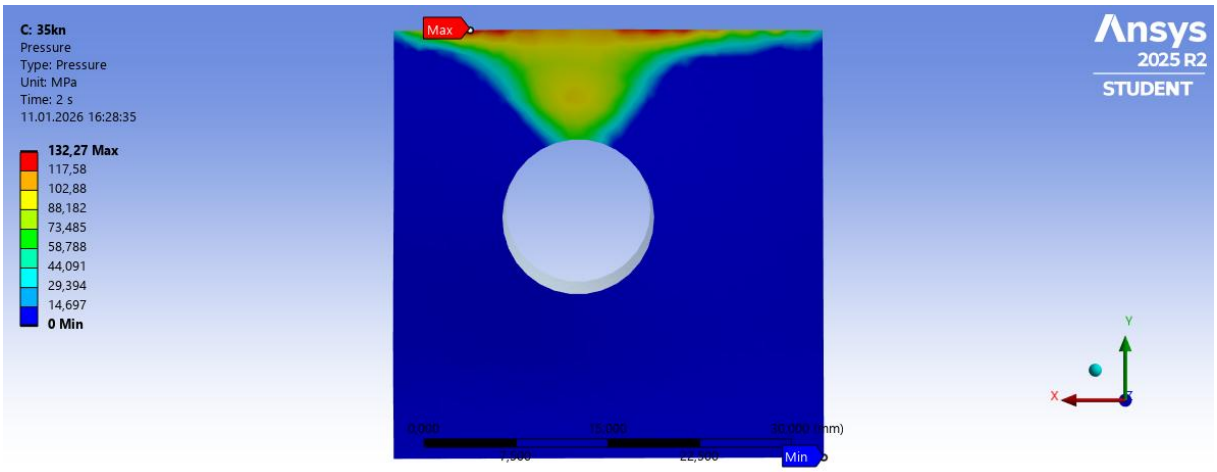


Figure 16: Contact pressure distribution at 35 kN external load showing reduced contact area.

At 40 kN, the joint approaches its critical limit. The contact pressure is confined to a very narrow ring around the bolt hole. The significant reduction in the blue/colored area confirms that the clamping force is being overcome by the external tension, and the joint stiffness is decreasing rapidly.

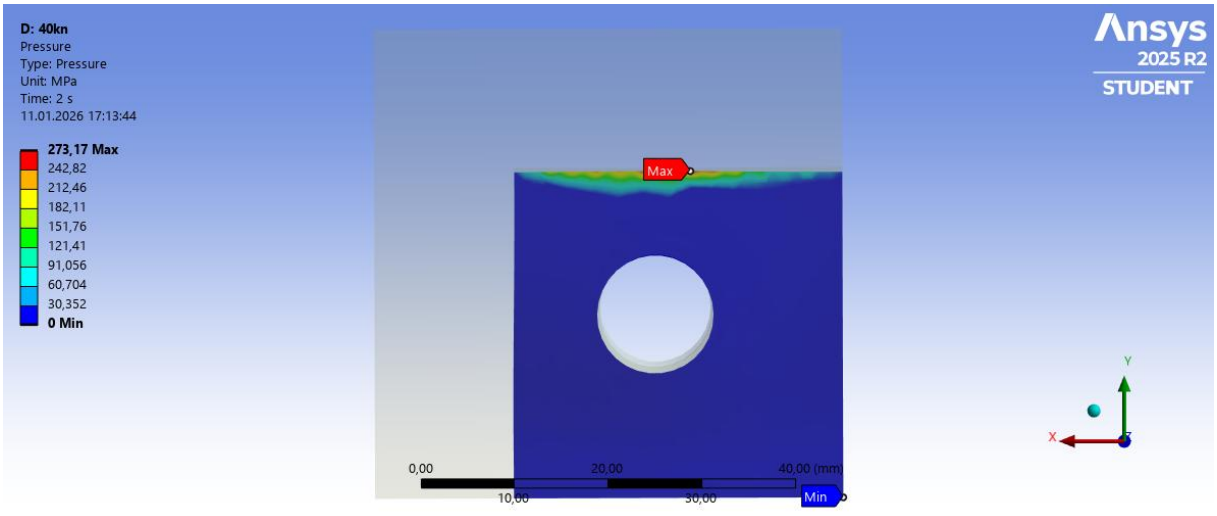


Figure 17: Contact pressure distribution at 40 kN.

The quantitative data presented in Table 2 was obtained through a parametric finite element simulation. The Applied Force served as the independent input variable, incrementally

increased within the Static Structural environment. For each load step, two key output parameters were monitored:

Bolt Working Load: Extracted using the Bolt Tool probe, representing the total axial tension acting on the bolt shank.

Average Contact Pressure: Calculated using the Contact Tool scoped to the frictional interface between the bracket and the base plate. The software computed the average pressure value across the active contact elements to monitor the separation status.

Applied Force (kN)	Bolt Working Load (N)	Average Contact Pressure (MPa)
25.00	45632	17.80
30.00	45647	16.51
35.00	45593	11.23
40.00	45644	7.34
43.00	45612	3.46
45.00	45591	0.74
45.53	45527	0.003

Table 2. Parametric study data summarizing the variation of Bolt Working Load and Average Contact Pressure under increasing external tensile force.

The variation of the Bolt Working Load as a function of the applied external force is presented in Figure 18.

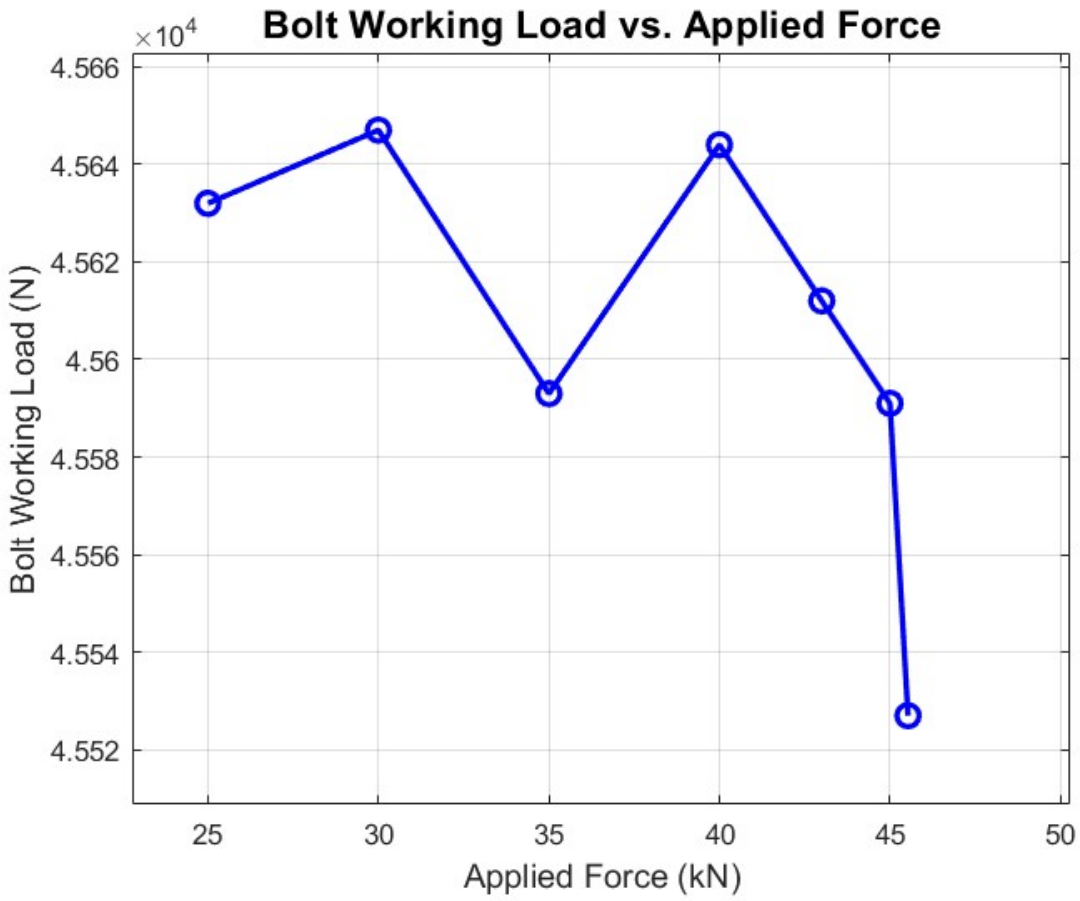


Figure 18: Variation of the bolt working load with increasing external force

As illustrated in the graph, the bolt working load remains remarkably constant (approx. 45.6 kN) throughout the loading range up to the separation point. This behavior is consistent with the mechanics of pre-tensioned joints with high stiffness. Before separation occurs, the external load is primarily counteracted by the relaxation of the compressive contact pressure between the plates, rather than by the stretching of the bolt. The bolt only experiences a fraction of the external load (defined by the joint stiffness factor, C), resulting in the observed flat response.

Conversely, the contact pressure at the bracket-plate interface exhibits a significant sensitivity to the external load. Figure 19 illustrates the linear decay of the average contact pressure.

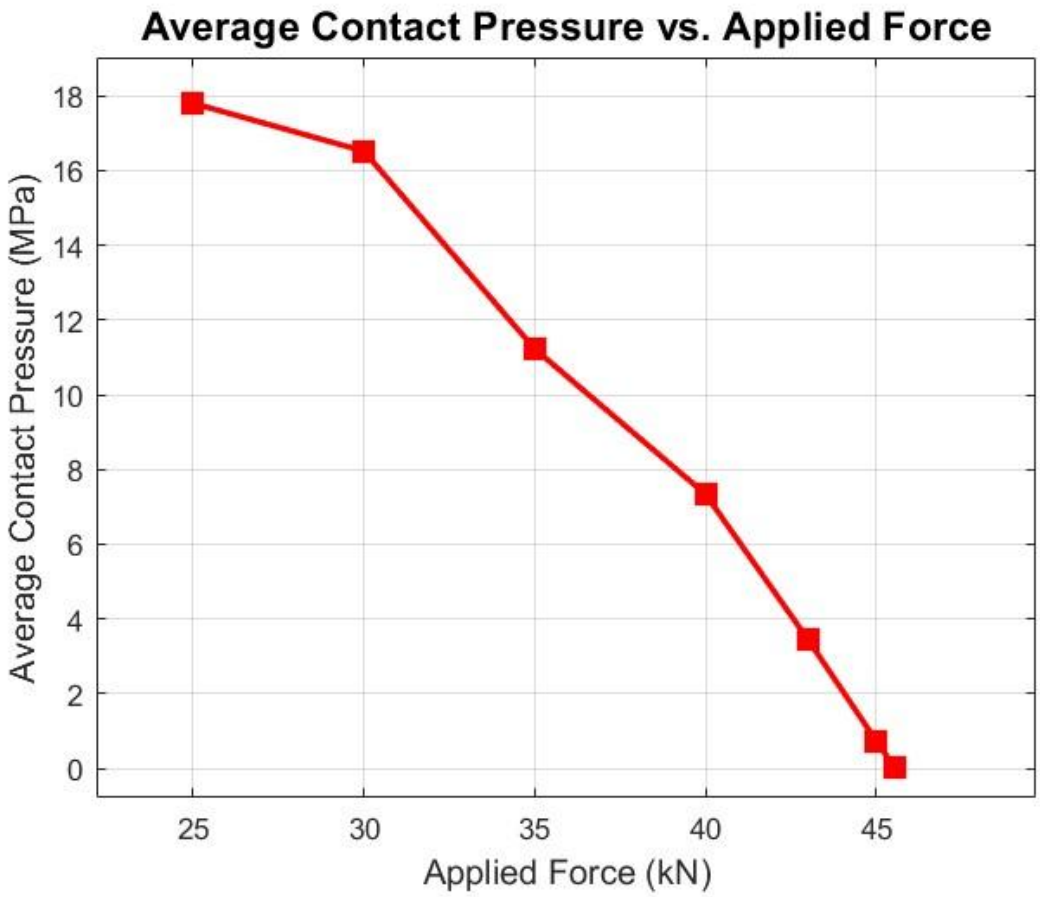


Figure 19: Reduction in average contact pressure as a function of external load.

The graph clearly demonstrates that the contact pressure decreases linearly as the external load increases. The intersection of this curve with the horizontal axis (zero pressure) marks the critical separation point. Based on the finite element results, the Separation Force is determined to be approximately 45.53 kN. At this load level, the external force completely overcomes the initial clamping force provided by the bolt pretension.

The physical separation is visually confirmed by the contact pressure contour at the near-separation load shown in Figure 20.

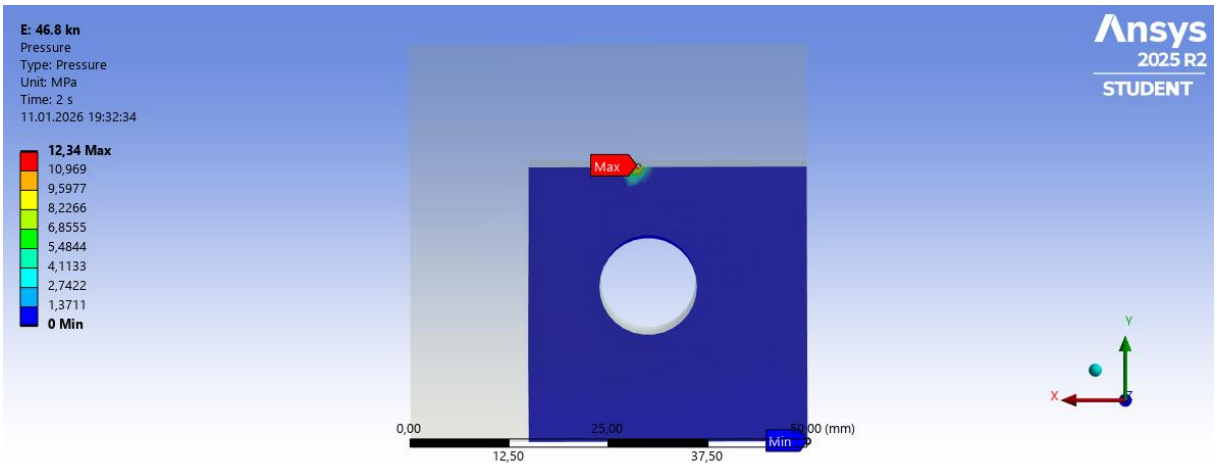


Figure 20: Contact pressure distribution near the separation limit, showing a significantly reduced contact area with average of 0.03 MPa.

Unlike the full contact observed at 25 kN (Task 1), the contact area at this high load is severely diminished, restricted to a thin region or vanishing entirely. This confirms that the joint has lost its structural rigidity and is on the verge of complete separation, validating the calculated separation force of 45.53 kN.

A visual inspection of the bolt working load at intermediate load steps reveals a remarkable consistency. As shown in the representative contour plot in Figure 21, both the magnitude and the distribution of the axial load along the bolt shank remain virtually unchanged despite the increase in external force. This phenomenon is a characteristic behavior of pre-tensioned joints with high stiffness; the external load is primarily absorbed by the relaxation of the compressive contact pressure between the clamped members rather than by the elongation of the bolt. Consequently, the bolt tension maintains its initial preload value until the separation point is reached.

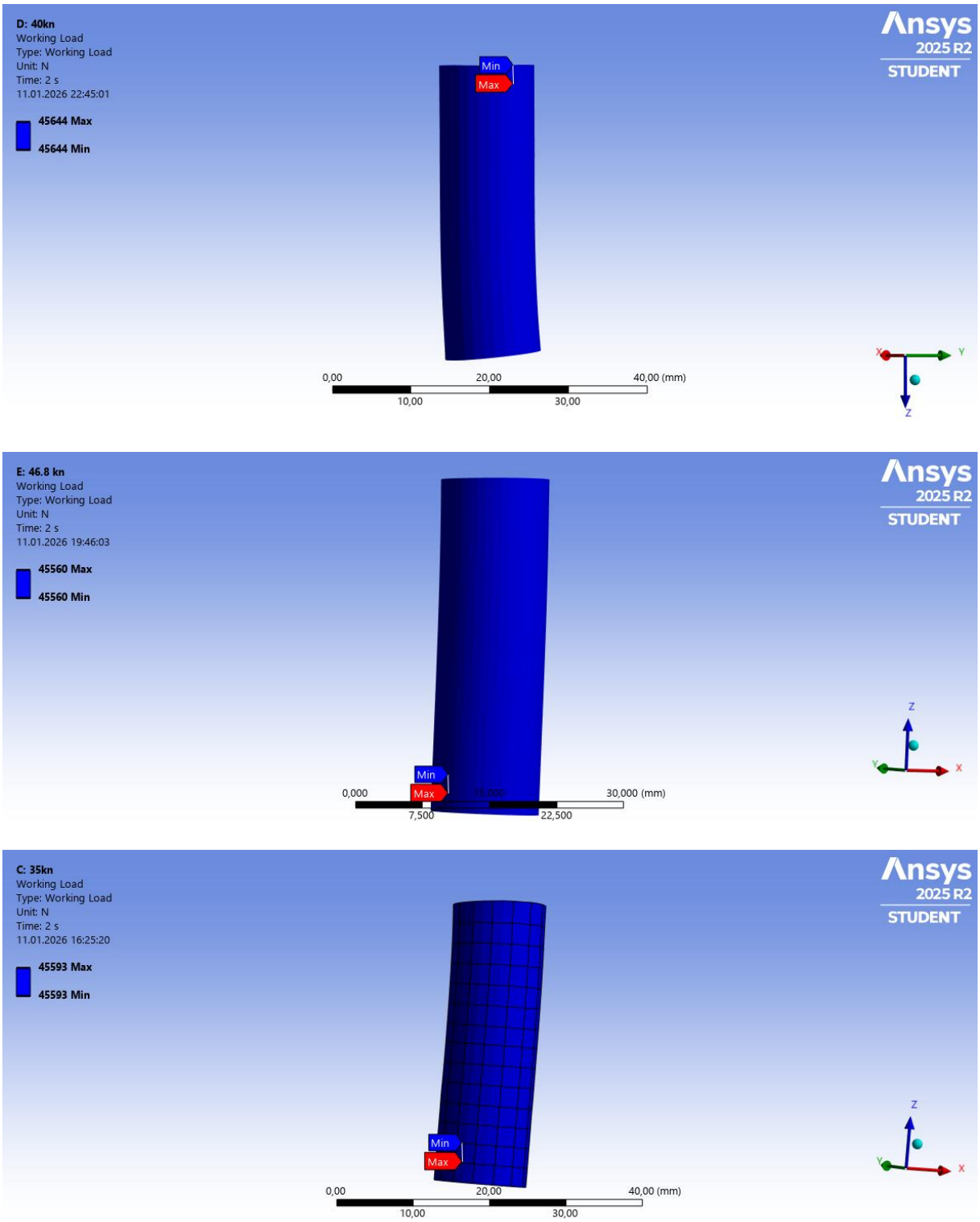


Figure 21: Working load remains nearly unchanged in every scenario.

3.3. Task 3: Design Optimization and Bolt Selection

In the final phase of the project, a design optimization study was conducted to propose a cost-effective and structurally sound solution for a reduced service load of 6000 N. The

original high-strength bolt configuration used in Task 1 was deemed over-designed for this lower load case. Therefore, the objective was to select a more economical bolt material and geometry while ensuring a Safety Factor (SF) greater than 1.2 against yielding and separation.

3.3.1. Design Decisions and Material Selection

To optimize the assembly, the following design changes were implemented:

- **Material Substitution:** The bolt material was switched from high-strength alloy steel to Metric Class 5.8 Carbon Steel. This material has a Yield Strength of 400 MPa and an Ultimate Tensile Strength of 500 MPa [3], which is sufficient for the target load and significantly reduces the component cost.
- **Preload Optimization:** A critical challenge with lower-grade bolts is their susceptibility to yielding during installation. The high pretension used in Task 1 (approx. 45 kN) would exceed the yield limit of a Class 5.8 bolt. Consequently, the pretension load was recalculated and optimized to 15,000 N. This value provides sufficient clamping force to prevent separation under the 6 kN external load while keeping the initial installation stresses well within the elastic range.

Specification	Value	Unit	Description
Bolt Standard	ISO 898-1	-	Metric Class 5.8 Carbon Steel
Nominal Diameter d	12	mm	Standard M12 Geometry
Thread Pitch p	1.75	mm	Coarse Thread

Specification	Value	Unit	Description
Nut Thickness t	10.8	mm	Standard M12 Nut Height (ISO 4032)
Tensile Stress Area A_t	84.3	Mm^2	Effective area for tension
Yield Strength S_y	400	MPa	Material Property
Ultimate Strength $S_{\{ut\}}$	500	MPa	Material Property

Table 3: Bolt and nut's specifications.

3.3.2. Analytical Verification

Before conducting the numerical simulation, analytical calculations were performed to estimate the safety factor based on the nominal stress area (A_t) of a standard M12 bolt ($84.3\ \mathrm{mm}^2$) and the selected Grade 5.8 properties.

First, the stress induced by the optimized preload ($F_i = 15{,}000\ \mathrm{N}$) was calculated:

$$\sigma_i = \frac{F_i}{A_t} = \frac{15{,}000}{84.3} \approx 142.3\ \mathrm{MPa}$$

Next, assuming a joint stiffness constant (C) of approximately 0.2, the additional stress component due to the external service load ($P = 6000\ \mathrm{N}$) was determined:

$$\Delta\sigma = \frac{C \times P}{A_t} = \frac{0.2 \times 6000}{84.3} \approx 14.2\ \mathrm{MPa}$$

The total theoretical tensile stress on the bolt is the sum of the preload stress and the external load component:

$$\sigma_{total} = \sigma_i + \Delta\sigma = 142.3 + 14.2 = 156.5 \text{ MPa}$$

Finally, the theoretical Safety Factor against yielding ($S_y = 400$ MPa) was calculated:

$$SF_{analytical} = \frac{S_y}{\sigma_{total}} = \frac{400}{156.5} \approx 2.55$$

It is important to note that this analytical value assumes a uniform stress distribution and does not account for stress concentrations at the thread roots or the head fillet. Therefore, the Finite Element Analysis (FEA) is expected to yield a lower, more conservative safety factor.

Calculate the shear stress (tau) on the threads using the project formula (assuming $t = 10$ mm for standard nut height):

$$\tau = \frac{F_b}{\pi \cdot d \cdot 0.75 \cdot t}$$

$$\tau = \frac{13{,}200}{\pi \cdot 12 \cdot 0.75 \cdot 10} = \frac{13{,}200}{282.74} = 46.7$$

Calculate Shear Yield Strength (S_{sy}) using Von-Mises criterion:

$$S_{sy} = \frac{S_y}{\sqrt{3}} = \frac{400}{1.732} = 230.9 \text{ MPa}$$

Calculate Safety Factor against Stripping ($SF_{stripping}$):

$$SF_{stripping} = \frac{S_{sy}}{\tau} = \frac{230.9 \text{ MPa}}{46.7 \text{ MPa}} = 4.94$$

Since $4.94 > 1.2$ the threads are safe against stripping

3.3.3. Verification and Safety Factor Results

The optimized configuration was simulated using the Finite Element Method to verify the structural integrity and capture the geometric stress concentrations. The distribution of the Safety Factor against yielding is presented in Figure 17.

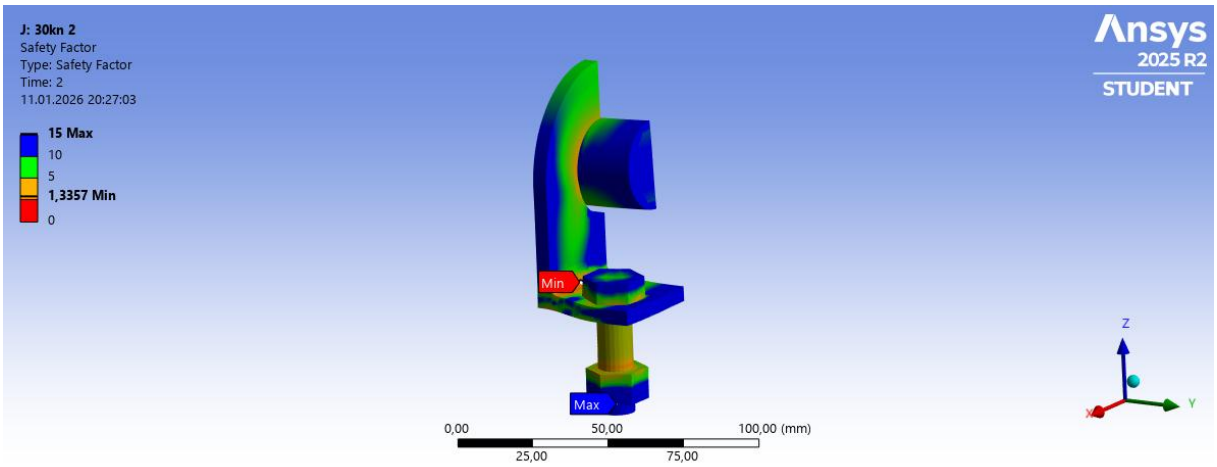


Figure 22: Safety Factor distribution on the optimized bolt (Class 5.8, 15 kN Preload) showing a minimum SF of 1.33.

The contour plot reveals that the minimum Safety Factor occurs at the stress concentration zones (thread root or head fillet) with a value of 1.3357.

3.3.4. FEA Results for Optimized Design

The contact pressure on the bracket's lower face is presented in Figure 23. The simulation confirms that a positive contact pressure is maintained around the bolt hole even under the external tensile load. The maximum pressure observed is approximately 85 MPa. The presence of this pressure ring proves that the clamping force is sufficient to prevent the bracket from separating from the plate.

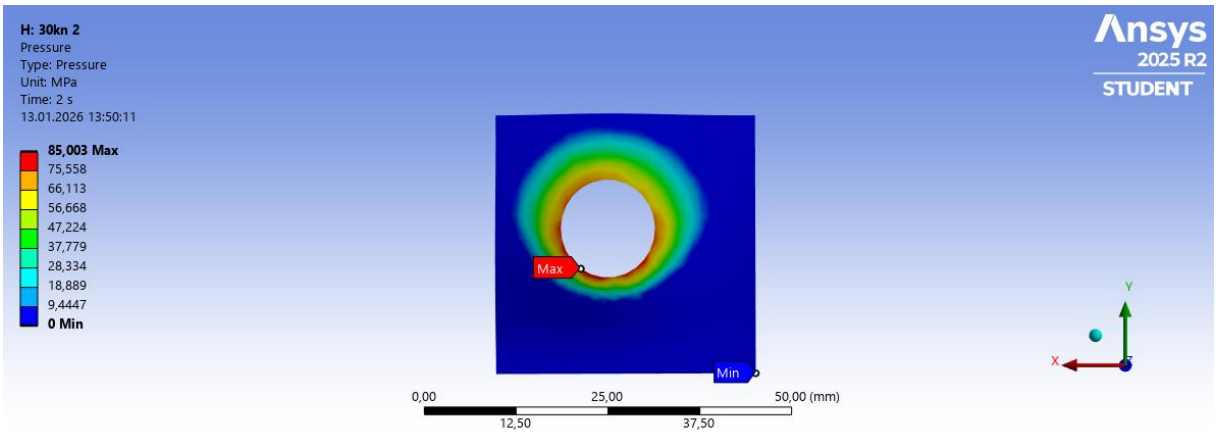


Figure 23: Contact Pressure distribution on the bracket's lower face under 6 kN service load.

The contact status of the interface is illustrated in Figure 24. The contact region is predominantly in the Sticking state (orange zone), indicating that there is no gap formation or significant sliding between the mating surfaces. This validates the analytical separation

check, confirming that the 15 kN preload provides an adequate safety margin against the 6 kN external force.

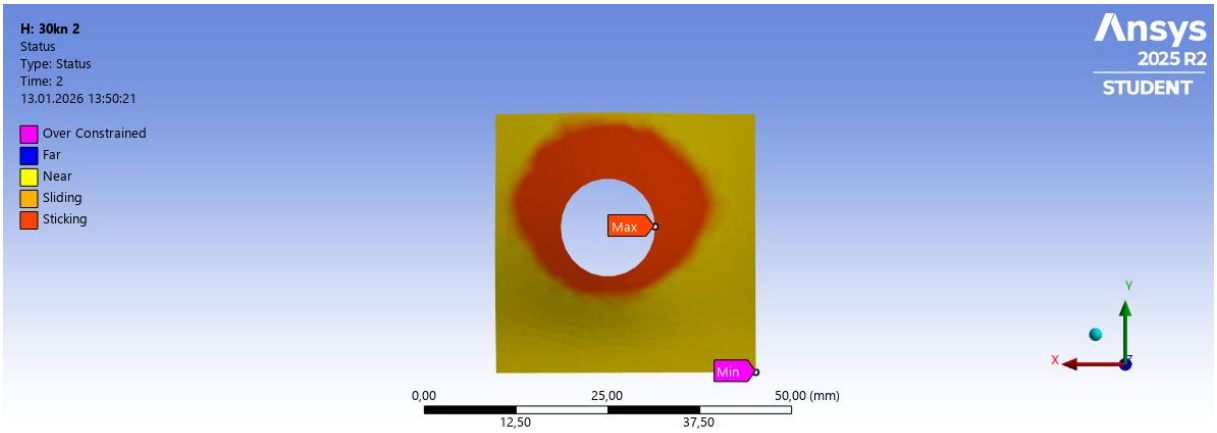


Figure 24: Contact Status indicating a fully closed Sticking joint.

Bolt Working Load The total axial force acting on the bolt (Working Load) during the application of the 6 kN external force is shown in Figure 25. The Finite Element Analysis predicts a total bolt load of 15,005 N.

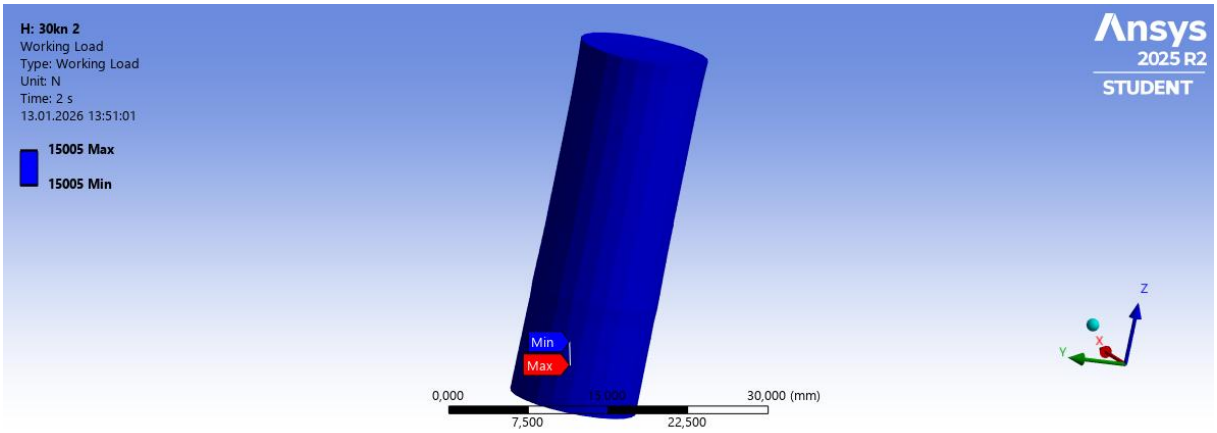


Figure 25: Bolt Working Load under 6 kN external force (approx. 15 kN).

4. Conclusion

In this project, a detailed finite element analysis was conducted to evaluate the structural behavior, separation limits, and design optimization of a single-bolt lap joint assembly. The study successfully utilized ANSYS Mechanical to simulate the complex interactions between bolt pretension, contact non-linearity, and external tensile loads.

The initial static analysis under a standard service load revealed that the stress distribution is highly localized. The maximum stresses were consistently observed at the geometric

discontinuities, specifically at the bolt head fillet and the inner radius of the bracket. Despite these concentrations, the joint maintained full structural integrity and adequate contact pressure to prevent separation under the 25 kN load.

The parametric study provided quantitative insights into the joint's stiffness and failure limits. It was observed that the bolt working load remains constant for a significant portion of the loading range, confirming that the external load is primarily absorbed by the relaxation of the contact pressure. The critical separation force was determined to be 45.53 kN, at which point the contact pressure dropped to zero, marking the functional failure of the joint.

Finally, the design optimization study demonstrated the potential for cost reduction without compromising safety. For a reduced service load of 6000 N, the high-strength bolt was successfully replaced with a Metric Class 5.8 carbon steel bolt. By optimizing the preload to 15,000 N, the design achieved a minimum Safety Factor of 1.34. This result not only satisfies the safety criteria but also prevents the common issue of bolt yielding during installation for lower-grade materials.

In conclusion, this study confirms that Finite Element Analysis is a vital tool for predicting bolted joint behavior. The results validate that proper material selection and preload optimization are essential for ensuring both the structural safety and the economic efficiency of mechanical assemblies.

5. References

[1] "Finite Element Analysis of a Bolted Lap Joint," LearnHub, Mech 303, Spring 2025.

[2] ANSYS® Academic Research Mechanical, Release 2025 R2, Help System, ANSYS, Inc.

[3] ISO 898-1:2013, *Mechanical properties of fasteners made of carbon steel and alloy steel – Part 1: Bolts, screws and studs with specified property classes*, International Organization for Standardization.